



Topic : Linux Files, Directories and Basic Commands (Part- 1)





Linux Files, Directories and Basic Commands

1. cd

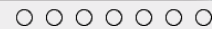
The **cd** command is used to change the current directory.

Syntax:

cd <directory name>

cd .. is used to change the directory one level up in our path.

```
shishir@shishir-VirtualBox: ~/NewFolder
shishir@shishir-VirtualBox:~$ pwd
/home/shishir
shishir@shishir-VirtualBox:~$ cd NewFolder
shishir@shishir-VirtualBox:~/NewFolder$ ls
demo.txt  NewFolder2  NewFolder3
shishir@shishir-VirtualBox:~/NewFolder$ cd NewFolder2
shishir@shishir-VirtualBox:~/NewFolder/NewFolder2$ cd ..
shishir@shishir-VirtualBox:~/NewFolder$
```





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2. ls

The **ls command** is used to display a list of the content of a directory.

Syntax:

ls

```
shishir@shishir-VirtualBox: ~  
shishir@shishir-VirtualBox:~$ ls  
Desktop    Downloads  'New Folder'  Public  Templates  
Documents  Music      Pictures      snap    Videos  
shishir@shishir-VirtualBox:~$
```



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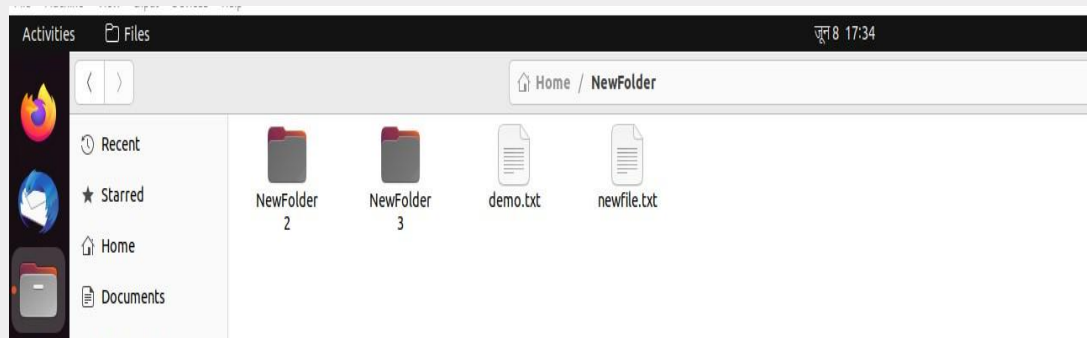
3. cp

The **cp** command is used to copy a file or directory.

Syntax:

cp <existing file name> <new file name>

```
shishir@shishir-VirtualBox: ~/NewFolder
shishir@shishir-VirtualBox:~/NewFolder$ ls
demo.txt  NewFolder2  NewFolder3
shishir@shishir-VirtualBox:~/NewFolder$ cp demo.txt newfile.txt
shishir@shishir-VirtualBox:~/NewFolder$
```





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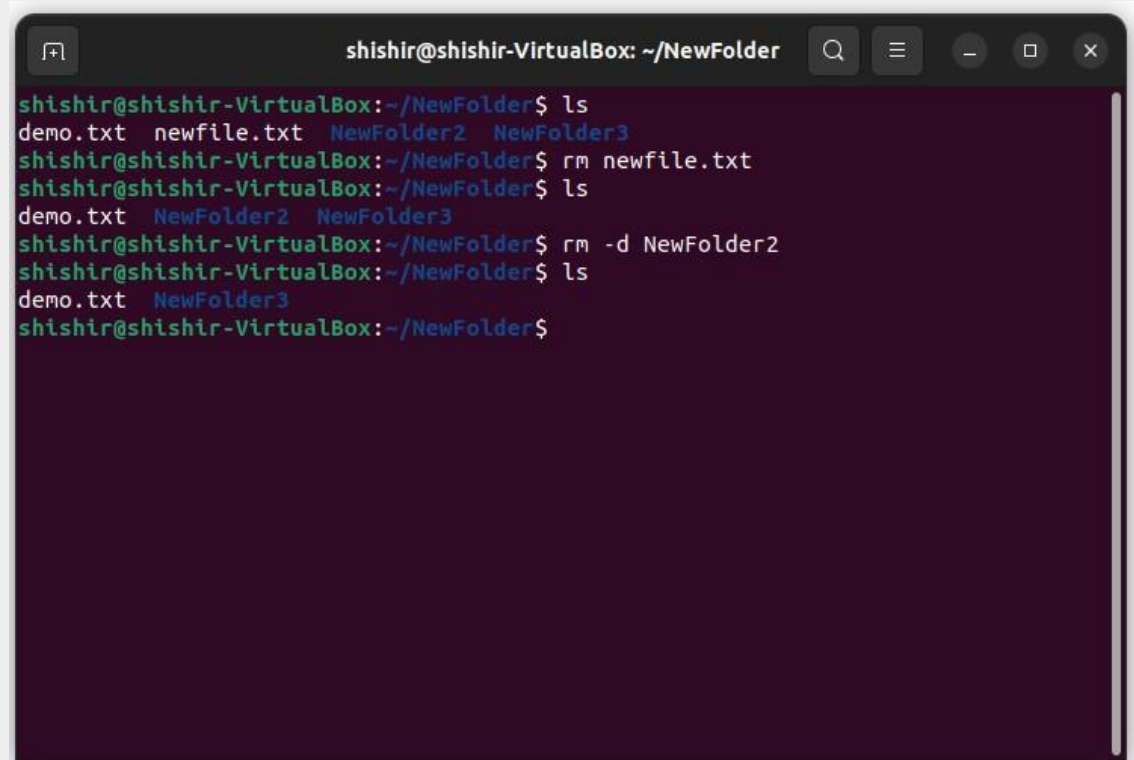
4. rm

The **rm** command is used to remove a file.

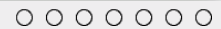
Syntax:

rm <file name>

We can also delete only empty directories using **-d flag**, otherwise we only use **rmdir** to remove directories.



```
shishir@shishir-VirtualBox: ~/NewFolder
shishir@shishir-VirtualBox:~/NewFolder$ ls
demo.txt  newfile.txt  NewFolder2  NewFolder3
shishir@shishir-VirtualBox:~/NewFolder$ rm newfile.txt
shishir@shishir-VirtualBox:~/NewFolder$ ls
demo.txt  NewFolder2  NewFolder3
shishir@shishir-VirtualBox:~/NewFolder$ rm -d NewFolder2
shishir@shishir-VirtualBox:~/NewFolder$ ls
demo.txt  NewFolder3
shishir@shishir-VirtualBox:~/NewFolder$
```





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5. mkdir

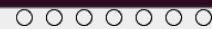
The **mkdir command** is used to create a new directory under any directory.

Syntax:

mkdir <directory name>

We can also create multiple directories as shown in the figure if we name multiple directories after the **mkdir command**.

```
shishir@shishir-VirtualBox: ~/NewFolder
shishir@shishir-VirtualBox:~/NewFolder$ ls
demo.txt  NewFolder3
shishir@shishir-VirtualBox:~/NewFolder$ mkdir NewFolder2
shishir@shishir-VirtualBox:~/NewFolder$ ls
demo.txt  NewFolder2  NewFolder3
shishir@shishir-VirtualBox:~/NewFolder$ mkdir NewFolder4 NewFolder5
shishir@shishir-VirtualBox:~/NewFolder$ ls
demo.txt  NewFolder2  NewFolder3  NewFolder4  NewFolder5
shishir@shishir-VirtualBox:~/NewFolder$
```



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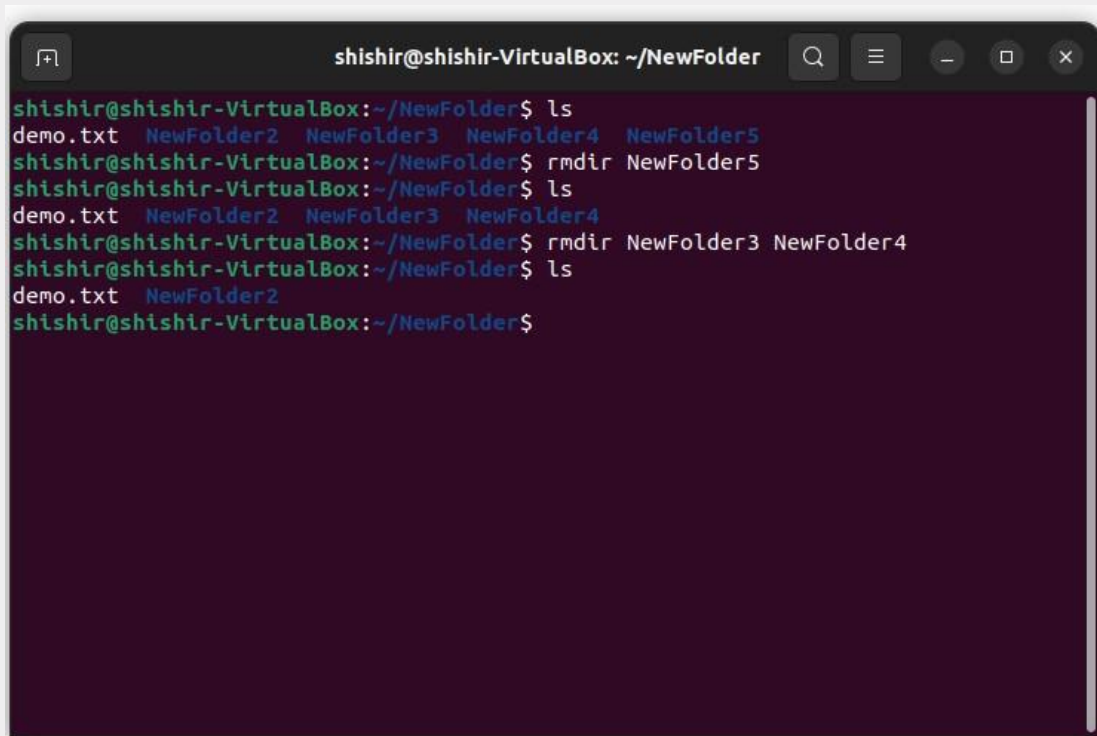
6. rmdir

The **rmdir** command is used to delete a directory.

Syntax:

rmdir <directory name>

We can also remove multiple directories as shown in the figure if we name multiple directories after the **rmdir** command.



```
shishir@shishir-VirtualBox: ~/NewFolder
shishir@shishir-VirtualBox:~/NewFolder$ ls
demo.txt  NewFolder2  NewFolder3  NewFolder4  NewFolder5
shishir@shishir-VirtualBox:~/NewFolder$ rmdir NewFolder5
shishir@shishir-VirtualBox:~/NewFolder$ ls
demo.txt  NewFolder2  NewFolder3  NewFolder4
shishir@shishir-VirtualBox:~/NewFolder$ rmdir NewFolder3 NewFolder4
shishir@shishir-VirtualBox:~/NewFolder$ ls
demo.txt  NewFolder2
shishir@shishir-VirtualBox:~/NewFolder$
```



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7. pwd

The **pwd** command is used to display the location of the current working directory.

Syntax:
pwd

```
shishir@shishir-VirtualBox: ~/NewFolder
shishir@shishir-VirtualBox:~/NewFolder$ pwd
/home/shishir/NewFolder
shishir@shishir-VirtualBox:~/NewFolder$
```





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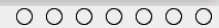
8. file

file command is used to determine the type of a file.

Syntax:

file <option> <filename>

```
shishir@shishir-VirtualBox: ~/NewFolder
shishir@shishir-VirtualBox:~/NewFolder$ ls
demo.txt  NewFolder2
shishir@shishir-VirtualBox:~/NewFolder$ file demo.txt
demo.txt: Unicode text, UTF-8 (with BOM) text
shishir@shishir-VirtualBox:~/NewFolder$
```





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9. cat

The **cat command** prints the contents of a specified file in the system's output.

Syntax:

cat <file name>

```
shishir@shishir-VirtualBox: ~/NewFolder
shishir@shishir-VirtualBox:~/NewFolder$ ls
demo.txt  newfile.txt  NewFolder2  NewFolder3
shishir@shishir-VirtualBox:~/NewFolder$ cat newfile.txt
This is a demo file written in OpenOffice by Shishir Ghimire at Ubuntu OS.
shishir@shishir-VirtualBox:~/NewFolder$
```



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10. more

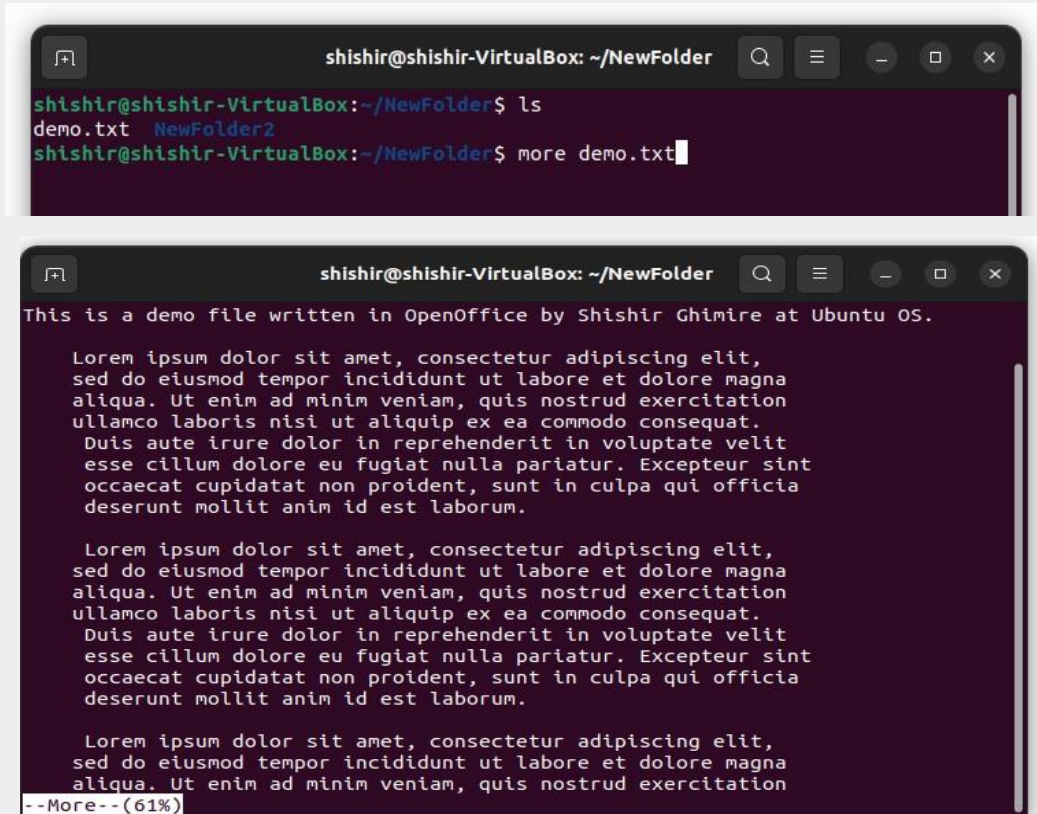
It works similarly as the **cat command** but the **more command** displays screenful output at a time in case of larger file (not the whole content).

Syntax:

more <file name>

In more command, the following keys are used to scroll the page:

1. **ENTER key:** To scroll down page by line.
2. **Space bar:** To move to the next page.
3. **b key:** To move to the previous page.
4. **/ key:** To search the string.



```
shishir@shishir-VirtualBox: ~/NewFolder
shishir@shishir-VirtualBox:~/NewFolder$ ls
demo.txt  NewFolder2
shishir@shishir-VirtualBox:~/NewFolder$ more demo.txt
This is a demo file written in OpenOffice by Shishir Ghimire at Ubuntu OS.

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aliqua. Ut enim ad minim veniam, quis nostrud exercitation
--More--(61%)
```

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11. less

It works similarly as the **more** command with some extra features such as adjustment in width and height of the terminal.

Syntax:

less <file name>

We have to press q to quit out of less.

```
shishir@shishir-VirtualBox: ~/NewFolder
shishir@shishir-VirtualBox:~/NewFolder$ ls
demo.txt  NewFolder2
shishir@shishir-VirtualBox:~/NewFolder$ less demo.txt
```

```
shishir@shishir-VirtualBox: ~/NewFolder
This is a demo file written in OpenOffice by Shishir Ghimire at Ubuntu OS.

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```

demo.txt

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12. chmod

The **chmod** command is used to change the access mode of a file.

Syntax:

chmod <options> <mode> <filename>

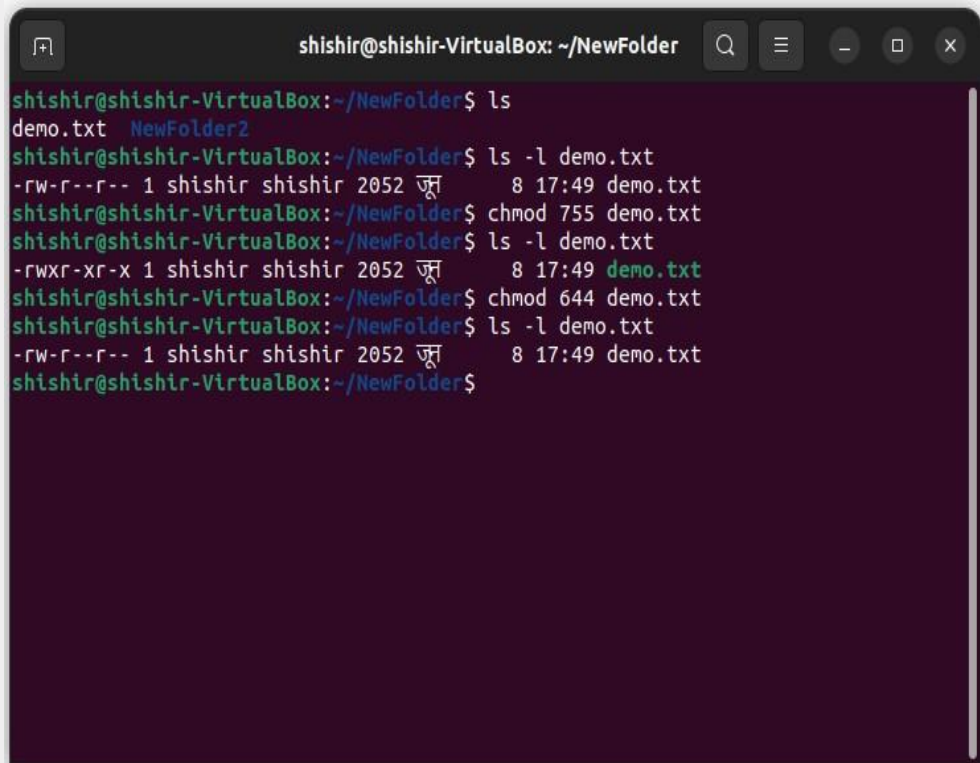
We can use the **chmod** command in 2 ways:

1. Symbol Mode:

Example: **chmod u+rw demo.txt**

2. Operator Mode:

Example: **chmod 755 demo.txt** as in figure.

A terminal window titled 'shishir@shishir-VirtualBox: ~/NewFolder' showing a series of commands and their outputs. The commands are: 'ls', 'ls -l demo.txt', 'chmod 755 demo.txt', 'ls -l demo.txt', 'chmod 644 demo.txt', and 'ls -l demo.txt'. The outputs show the file permissions changing from '-rw-r--r--' to '-rwxr-xr-x' after the first chmod command, and back to '-rw-r--r--' after the second chmod command. The file 'demo.txt' is located in 'NewFolder2'.

```
shishir@shishir-VirtualBox: ~/NewFolder$ ls
demo.txt  NewFolder2
shishir@shishir-VirtualBox: ~/NewFolder$ ls -l demo.txt
-rw-r--r-- 1 shishir shishir 2052 जून  8 17:49 demo.txt
shishir@shishir-VirtualBox: ~/NewFolder$ chmod 755 demo.txt
shishir@shishir-VirtualBox: ~/NewFolder$ ls -l demo.txt
-rwxr-xr-x 1 shishir shishir 2052 जून  8 17:49 demo.txt
shishir@shishir-VirtualBox: ~/NewFolder$ chmod 644 demo.txt
shishir@shishir-VirtualBox: ~/NewFolder$ ls -l demo.txt
-rw-r--r-- 1 shishir shishir 2052 जून  8 17:49 demo.txt
shishir@shishir-VirtualBox: ~/NewFolder$
```

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13. chown

The **chown command** is used to change a file's ownership, directory, or symbolic link for a user or a group.

1. To change the owner:

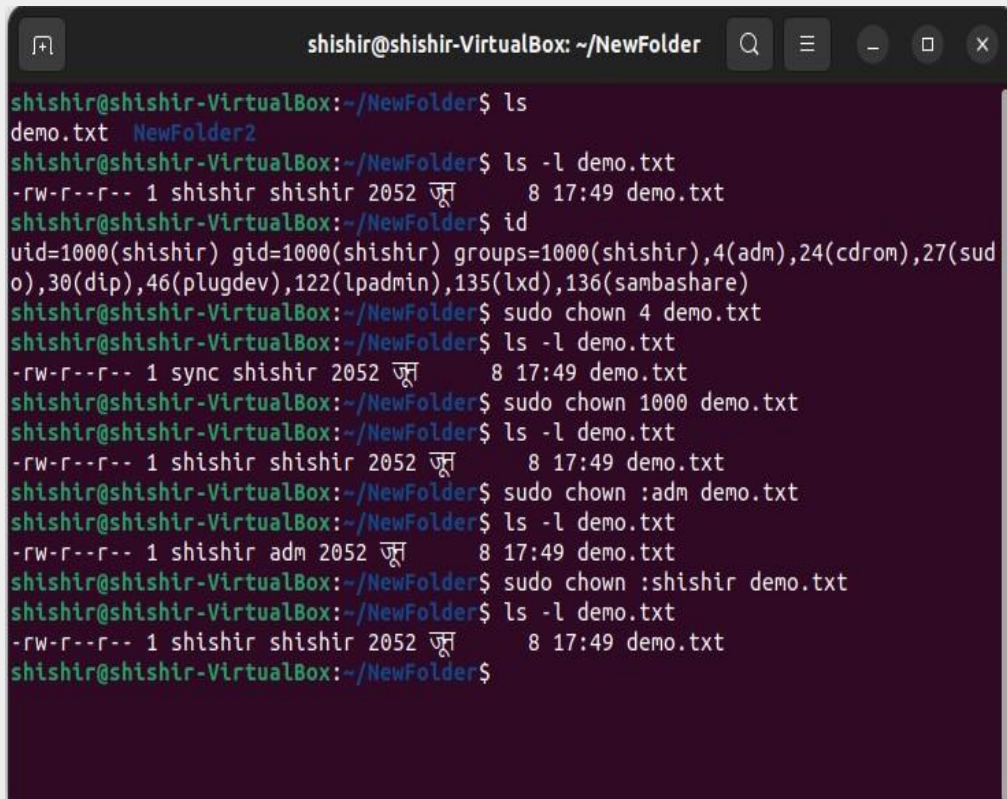
sudo chown <username> <File name>

Example: **sudo chown 1000 demo.txt** as in the figure.

2. To change owner and group:

**sudo chown <username/UID> : <Groupname/GID>
<filename>**

Example: **sudo chown : shishir demo.txt** as in figure.



```
shishir@shishir-VirtualBox: ~/NewFolder
shishir@shishir-VirtualBox:~/NewFolder$ ls
demo.txt  NewFolder2
shishir@shishir-VirtualBox:~/NewFolder$ ls -l demo.txt
-rw-r--r-- 1 shishir shishir 2052 जू  8 17:49 demo.txt
shishir@shishir-VirtualBox:~/NewFolder$ id
uid=1000(shishir) gid=1000(shishir) groups=1000(shishir),4(adm),24(cdrom),27(sudo),30(dip),46(plugdev),122(lpadmin),135(lxd),136(sambashare)
shishir@shishir-VirtualBox:~/NewFolder$ sudo chown 4 demo.txt
shishir@shishir-VirtualBox:~/NewFolder$ ls -l demo.txt
-rw-r--r-- 1 sync shishir 2052 जू  8 17:49 demo.txt
shishir@shishir-VirtualBox:~/NewFolder$ sudo chown 1000 demo.txt
shishir@shishir-VirtualBox:~/NewFolder$ ls -l demo.txt
-rw-r--r-- 1 shishir shishir 2052 जू  8 17:49 demo.txt
shishir@shishir-VirtualBox:~/NewFolder$ sudo chown :adm demo.txt
shishir@shishir-VirtualBox:~/NewFolder$ ls -l demo.txt
-rw-r--r-- 1 shishir adm 2052 जू  8 17:49 demo.txt
shishir@shishir-VirtualBox:~/NewFolder$ sudo chown :shishir demo.txt
shishir@shishir-VirtualBox:~/NewFolder$ ls -l demo.txt
-rw-r--r-- 1 shishir shishir 2052 जू  8 17:49 demo.txt
shishir@shishir-VirtualBox:~/NewFolder$
```



ThankYou!!!

